



Islamic University of Technology
Department of Computer Science and Engineering

Lab 10: Max Flow

CSE 4404: Algorithms Lab
Summer 2023-24

Task A. Download Speed

Time Limit: 2 seconds | Memory Limit: 512 MB

Consider a network consisting of n computers and m connections. Each connection specifies how fast a computer can send data to another computer. Kotivalo wants to download some data from a server. What is the maximum speed he can do this, using the connections in the network?

Input Format

The first input line has two integers n and m : the number of computers and connections. The computers are numbered $1, 2, \dots, n$. Computer 1 is the server and computer n is Kotivalo's computer. After this, there are m lines describing the connections. Each line has three integers a, b and c : computer a can send data to computer b at speed c .

Output Format

Print one integer: the maximum speed Kotivalo can download data.

Constraints

- $1 \leq n \leq 500$
- $1 \leq m \leq 1000$
- $1 \leq a, b \leq n$
- $1 \leq c \leq 10^9$

Examples

Sample Input	Sample Output
4 5 1 2 3 2 4 2 1 3 4 3 4 5 4 1 3	6

Task B. Distinct Routes

Time Limit: 2 seconds | Memory Limit: 512 MB

A game consists of n rooms and m teleporters. At the beginning of each day, you start in room 1 and you have to reach room n . You can use each teleporter at most once during the game. How many days can you play if you choose your routes optimally?

Input Format

The first input line has two integers n and m : the number of rooms and teleporters. The rooms are numbered $1, 2, \dots, n$. After this, there are m lines describing the teleporters. Each line has two integers a and b : there is a teleporter from room a to room b . There are no two teleporters whose starting and ending room are the same.

Output Format

Print an integer k : the maximum number of days you can play the game.

Constraints

- $2 \leq n \leq 500$
- $1 \leq m \leq 1000$
- $1 \leq a, b \leq n$

Examples

Sample Input	Sample Output
6 7 1 2 1 3 2 6 3 4 3 5 4 6 5 6	2

Marks Distribution

Task	Marks
Task A	50%
Task B	50%